

Controlling the dispersion of multi-wall carbon nanotubes in aqueous surfactant solution

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Abstract

The sonication-driven dispersion of multi-wall carbon nanotubes (MWCNTs) in aqueous surfactant solution has been monitored by UV–vis spectroscopy and transmission electron microscopy. Time dependent sonication experiments reveal that the maximum achievable dispersion of MWCNTs corresponds to the maximum UV–vis absorbance of the solution. With higher surfactant concentration the dispersion rate of MWCNTs increases and less total sonication energy is required to achieve maximum dispersion. Dispersion of higher MWCNT concentrations requires higher total sonication energy. For effective dispersion the minimum weight ratio of surfactant to MWCNTs is 1.5–1. The surfactant molecules are adsorbed on the surface of the MWCNTs and prevent re-aggregation of MWCNTs so that a colloidal stability of MWCNT dispersions could be maintained for several months. The maximum concentration of MWCNTs that can be homogeneously dispersed in aqueous solution is about 1.4 wt%.

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1. Introduction

Since the study published by Iijima in 1991 [1] carbon nanotubes (CNTs) have attracted considerable attention of researchers all over the world because of their unique electronic, thermal, optical and mechanical properties [2,3]. Among CNTs, single-wall carbon nanotubes (SWCNTs) consist of a single layer of graphite lattice rolled into a perfect cylinder with a diameter usually in the range of 0.7–2 nm, whereas sets of concentric cylindrical graphite shells form multi-wall carbon nanotubes (MWCNTs). The aspect ratios of both SWCNTs and MWCNTs are very high since their lengths are in the range of several micrometers [4]. SWCNTs possess elastic modu-

lus and breaking strength of about 1 TPa and 37 GPa, respectively [5,6], and they can be theoretically described as “one-dimensional” conductors due to the delocalization of the π -electrons along the nanotube axis. Properties of MWCNTs are less superior, however, excellent mechanical and electrical properties are reported [7,8].

Because CNTs as produced invariably exist as aggregates or bundles that are tightly bound by an estimated interaction of 500 eV/ μ m of tube length for SWCNT [9,10], one of the key challenges is processing or engineering CNTs for potential applications such as electronic components, reinforced coatings or conductive polymer composites. Significant research efforts have already been devoted over the last years to achieve this goal [11–16]. The main route available to engineering CNTs is solution based processing [15]. In particular, surfactants such as sodium dodecyl sulphate (SDS) have been studied for their excellent nanotube stabilization and separation capabilities [16–18]. It has been proposed that SWCNTs are either

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encapsulated in a cylindrical micelle, or coated by hemimicelles, but the exact mechanism has not been conclusively established yet [19,20]. The dispersion of CNTs in solution can be studied by cryogenic transmission electron microscopy (cryo-TEM) [21], atomic force microscopy (AFM) [22], scanning electron microscopy [23], Raman spectroscopy [24], and optical microscopy [25]. Moreover, the discovery of nanotube fluorescence [17] offers a more precise method for detecting dispersion of individual nanotubes, and UV–visible spectroscopy has been successfully used to monitor the exfoliation dynamics of SWCNTs in aqueous solution [26]. Though, systematic investigation on parameters determining the dispersion behaviour of CNTs, such as sonication energy, applied concentration of CNTs, ratio of CNTs to surfactant and so on, has not yet been performed. It is the purpose of our study to present details on the dispersion mechanism of MWCNTs in aqueous surfactant solution and on the factors that influence its efficiency. The dispersion of the MWCNTs is studied by using UV–vis spectroscopy and TEM.

2. Experimental

MWCNTs used were purified thin MWCNTs (batch number MWA P 041206) provided by Nanocyl Belgium. The surfactant used for the dispersion of the MWCNTs was sodium dodecyl sulfate (SDS; 90%) provided by Merck Chemical Co. All dispersion experiments were carried out with distilled water. All solutions were prepared by mixing a certain amount of MWCNTs with 20 mL of an aqueous SDS solution in a flask, after which the resulting mixture was sonicated for different times under mild conditions. All sonication processes were carried out with a horn sonicator (Sonic Vibracell VC750) with a cylindrical tip (10 mm end cap diameter). The output power was fixed at 20 W, thus delivering energy of 1100–1200 J/min. The flask was placed in a bath of ice water during sonication in order to prevent rising of the temperature of the mixture.

UV–vis absorption spectra were recorded with a Hewlett-Packard 8453 spectrometer operating between 200 and 1100 nm. Samples were taken regularly during the sonicating process and diluted by a certain factor, resulting in certain MWCNT contents that were suitable for UV–vis measurements. The blank used was the original SDS solution diluted by the same factor, under the same conditions as the samples themselves. Transmission electron microscopy (TEM) observations of MWCNT dispersions were performed using a Tecnai 20 (FEI Co.) operated at 200 kV. The samples were prepared by dipping a copper TEM grid in the MWCNT dispersion and subsequent drying.

3. Results and discussion

3.1. UV–vis spectra of MWCNTs–SDS solutions

Individual CNTs are active in the UV–vis region and exhibit characteristic bands corresponding to additional absorption due to 1D van Hove singularities [27–29]. Bundled CNTs, however, are hardly active in the wavelength region between 200 and 1200 nm – their photoluminescence is quenched – most probably because of carrier are tunneling between the nanotubes [30]. Therefore, it is possible to establish a relationship between the amount of CNTs individually dispersed in solution and the intensity of the corresponding absorption spectrum [11]. Moreover, UV–vis

spectroscopy can be used to monitor the dynamics of this dispersion process of CNTs, allowing the determination of the optimal sonicating time [25].

Fig. 1 illustrates UV–vis spectra of MWCNT–SDS solutions, obtained after different sonication times. After sonication the absorbance of MWCNT solutions shows a maximum between 200 and 300 nm and gradually decreases from UV to near-IR, which is partly due to scattering, especially in the lower wavelength range. Similar results are reported for UV–vis absorption spectra of SWCNTs by Jiang et al. [11] and Grossiord et al. [26]. At the beginning of sonication, MWCNTs exist as big aggregates and bundles in solution that are strongly entangled, and no absorption is evident in the UV–vis spectrum (Fig. 1a). During sonication, the provided mechanical energy can indeed overcome the van der Waals interactions in the MWCNTs bundles and lead to their disentanglement and dispersion. The increasing amount of dispersed MWCNTs results in an increasing area below the spectrum lines representing the absorbance (Fig. 1b–g).

3.2. Effect of SDS on the dispersion process of MWCNTs in solution

The power of sonication cannot be kept constant throughout the experiments due to the voltage fluctuation, and the mechanical energy delivered by sonication is related to the sonication power and time. Thus, it is advisable to plot the absorbance versus the total energy supplied to the solution instead of sonication time, in order to monitor the dispersion dynamics of MWCNTs in solution. Moreover, since we verified that during the sonication process the relative evolution of the area under the spectrum is proportional to the relative absorbance value at a specific wavelength, it was decided to determine the absorbance maximum around 260 nm and to plot this value as a

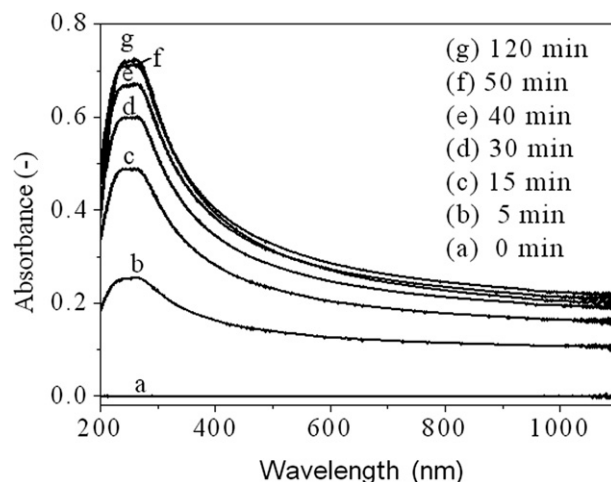


Fig. 1. Evolution of UV–vis spectra of an aqueous 0.1 wt% MWCNT–0.15 wt% SDS solution as a function of sonication time at continuous power of 20 W (solutions are diluted by a factor of 150).

function of the total energy supplied to the solution. All absorbance intensities are used after baseline subtraction.

Fig. 2 shows the evolution of the maximum absorbance as a function of the total energy supplied to the solution, describing the dispersion dynamics of 0.1 wt% MWCNTs in aqueous solutions with different concentrations of SDS. The general trends of the UV absorbance versus total energy curves obtained for all MWCNT–SDS solutions are very similar. After an increase at the beginning of the sonication process, the value of maximum absorbance reaches a plateau value after a certain amount of supplied energy, which corresponds to the maximum achievable degree of dispersion of the MWCNTs in the aqueous SDS solutions. With higher SDS concentration the plateau value increases, the dispersion rate of MWCNTs is higher at the beginning of the sonication process, and the total energy required achieving the maximum dispersion of MWCNTs decreases (Fig. 2). It indicates that the SDS molecules can exert an “unzipping force” to disperse MWCNTs in the solution, which is supported by the high local shear provided by sonication. Evidence for the role of surfactant in the dispersion of SWCNTs during sonication has been reported by Strano et al. [18]. During the sonication process, the bundle ends of SWCNTs are “frayed” by high local shear and become the site for additional SDS adsorption; then the SDS molecules gradually exfoliate the SWCNTs bundles in an “unzipping” mechanism. In case of MWCNTs a similar process can be anticipated. However, quantification of the dispersion state of MWCNTs by means of UV–vis still is challenging.

In order to visualize the dispersion state of MWCNTs in solution and to support the interpretation of the UV–vis results, TEM investigations have been performed. Fig. 3 shows TEM bright-field images of MWCNTs, which were prepared from an aqueous SDS solution after different sonication times (the weight ratio of SDS to MWCNTs is 1.5:1). After 5 min of sonication, which corresponds to a total sonication energy of 6000 J, the MWCNTs mainly

remain as big aggregates, and only a small amount of MWCNTs is exfoliated (Fig. 3a). This corresponds to the low absorption in the UV–vis spectrum, as shown in Fig. 1b. However, after 90 min of sonication (corresponding to a total sonication energy of 100000 J), hardly any big nanotube aggregates can be seen in the TEM image (Fig. 3b), indicating that almost all the MWCNTs are exfoliated after the maximum achievable dispersion has been reached. Several other samples have been investigated, being prepared in a similar fashion, and all demonstrate a comparable appearance of the MWCNT dispersion; the TEM observations are consistent with the UV–vis results.

Since the surfactant molecules adsorbed on the MWCNT surface may block off charge transport through a MWCNT network, which is important for e.g. electronic applications, the minimum amount of SDS should be determined that still guarantees maximum dispersion of the MWCNTs in aqueous solution. From Fig. 2 it can be seen that dispersion evolution of 0.1 wt% MWCNTs in solution proceeds similar for SDS concentrations of 0.15 wt% or higher, indicating that the optimum ratio of SDS to MWCNTs is 1.5:1 by weight. The utmost dispersion of MWCNTs in solution cannot be achieved at lower SDS/MWCNT ratios, because sufficient surfactant is required to produce an efficient coating on the surface of the MWCNTs to prevent re-aggregation [16].

Fig. 4 shows TEM bright-field images of individual MWCNTs coated with a layer of adsorbed SDS. The MWCNTs used have outside diameters of about 10–20 nm, and the thickness of the adsorbed SDS layer is about 5–10 nm. From the high resolution TEM image of Fig. 4b, it can be discerned that the MWCNTs are coated with a layer of surfactant that has a thickness of a few nanometers. Most probably, this layer is not a monolayer of SDS surfactant. No further details can be observed on the internal organization of the surfactant layer. Cryo-TEM investigations might provide more information about MWCNT/SDS interactions in solution.

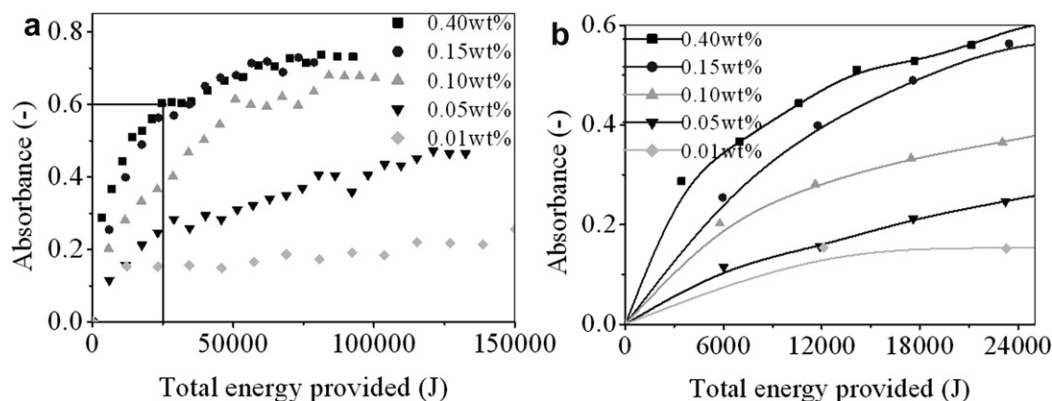


Fig. 2. (a) Evolution of the maximum absorbance of 0.1 wt% MWCNTs in aqueous solution for different SDS concentrations; (b) zoom-in of the curve at the beginning of the sonication process to show the dispersion rates for different SDS concentrations. The sonicated solutions are diluted by a factor of 150 when taking UV–vis measurements.

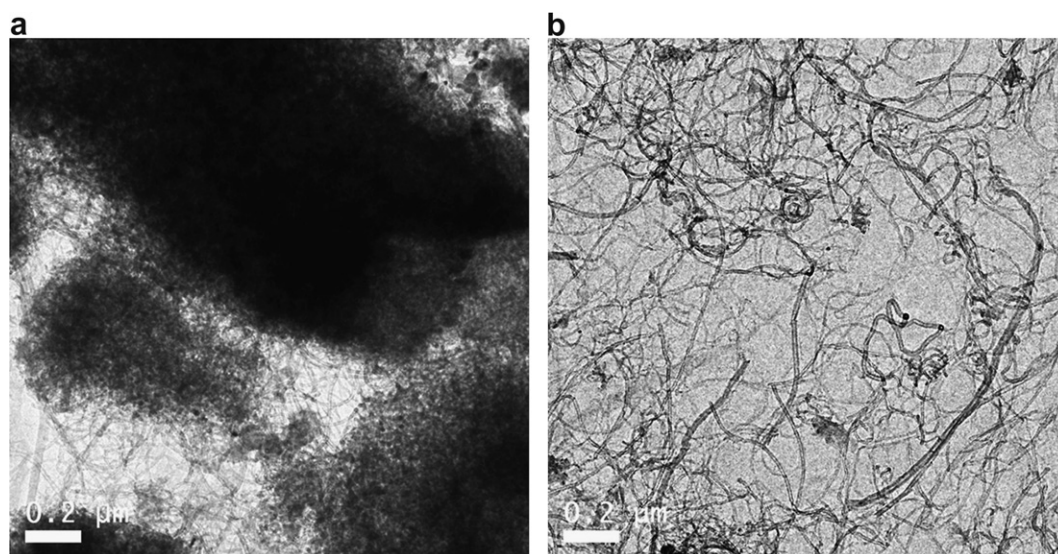


Fig. 3. TEM images of MWCNTs from an aqueous 0.1 wt% MWCNT–0.15 wt% SDS solution after different sonication times at 20 W: (a) 5 min, corresponding to total sonication energy of 6000 J; (b) 90 min, corresponding to total sonication energy of 100 000 J. The solution has been diluted 10 times before casting.

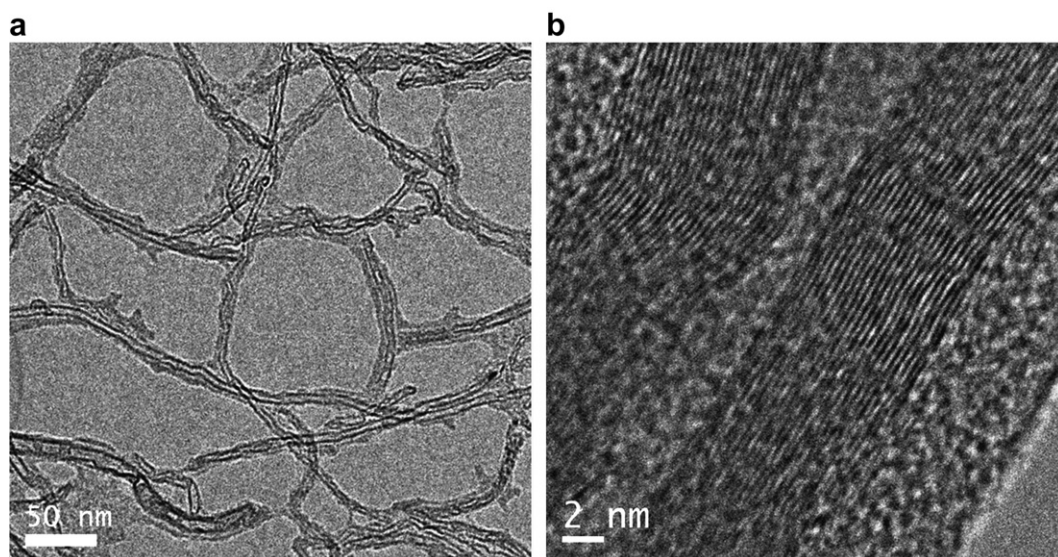


Fig. 4. TEM images of SDS-coated (a) individual MWCNTs and (b) single MWCNT at higher resolution. The weight ratio of SDS to MWCNTs in the solution was 1.5:1.

3.3. Dispersion of high concentrations of MWCNTs in solution

For effective processing dispersions with high concentrations of exfoliated MWCNTs are required. Fig. 5 shows the dispersion dynamics of different concentrations of MWCNTs in aqueous SDS solution. For each UV–vis measurement the absolute amount of MWCNTs was made identical by applying the appropriate dilution factor, thus the observed absorbance represents the state of dispersion. Although the shapes as well as the plateau values of the UV absorbance versus total sonication energy curves are very

similar for different concentrations of MWCNTs, there is one evident difference: only 60 000 J is necessary to reach the maximum degree of dispersion for 0.1 wt% MWCNTs in solution, whereas almost 210 000 J is required for 1.4 wt% MWCNTs to achieve the same state of dispersion. Thus, more energy is required to disperse a higher amount of MWCNTs in solution.

However, when the MWCNT concentration reaches 1.5 wt%, sonication does not result in efficient dispersion, not even after very long sonication time, or corresponding sonication energy of about 270 000 J. Such a behavior can presumably be ascribed to the reduction of electrostatic

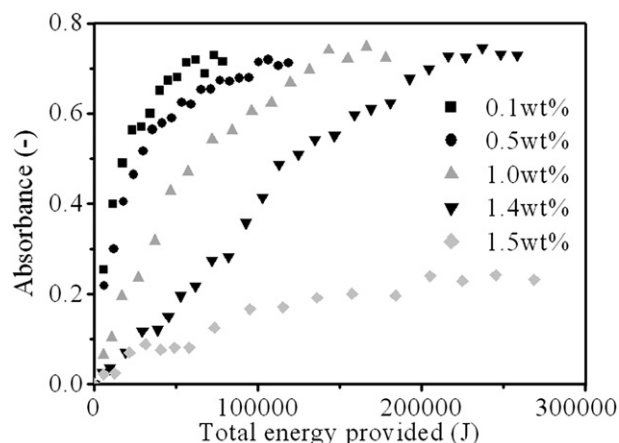


Fig. 5. Dispersion evolution of different concentrations of MWCNTs in aqueous solution as a function of the energy delivered to the solution. The weight ratio of SDS to MWCNTs in solution is 1.5:1. Before the UV–vis measurements, the solutions were diluted by different factors, resulting in the same MWCNTs content; the blank used is the original SDS solution diluted by the same factor.

repulsion forces between MWCNTs due to the forming of a too large amount of SDS micelles in the aqueous solution. The osmotic pressure of micelles around the MWCNT bundles creates an effective attraction, resulting in the depletion-induced aggregation of MWCNTs. This behavior also is mentioned in Ref. [25] as a possible reason for aggregation of SWCNTs.

It is worthwhile to mention that there is a minimum energy required to maximally disperse a certain amount of MWCNTs in aqueous solution, which can be determined by using UV–vis spectroscopy. Prolonged sonication is not only energy-consuming, it will also damage and cut CNTs [17,22], which should be avoided since it is often advantageous to use the high aspect ratio of CNTs in various applications. Therefore, the sonicating process should be stopped when the ultimate dispersion of the CNTs has been achieved.

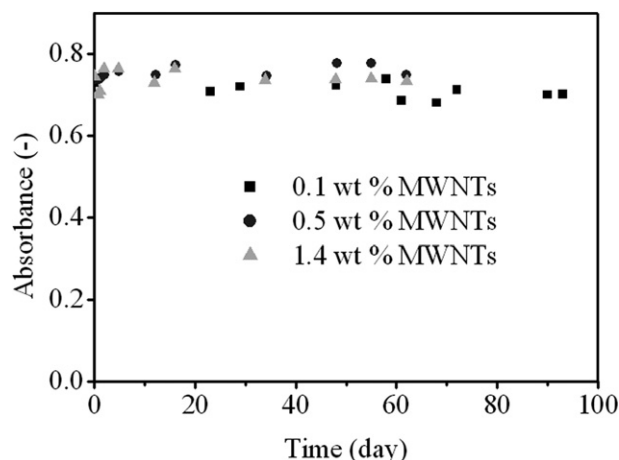


Fig. 6. Colloidal stability of the dispersed MWCNTs in aqueous solution. The weight ratio of SDS to MWCNTs in solution is 1.5:1.

3.4. Colloidal stability of MWCNTs in solution

During sonication SDS helps to exfoliate and disperse MWCNTs, and afterwards SDS molecules are adsorbed on the surface of MWCNTs (Fig. 4) and prevent aggregation of MWCNTs by electrostatic repulsion [16]. Long term stability of MWCNT dispersions is demonstrated by UV–vis measurements of dispersions stored for various times. During the observation time frame, the maximum UV–vis absorbance of the dispersions remains almost unchanged during 90 days (Fig. 6). Therefore, the colloidal stability of these dispersions is maintained for several months.

4. Conclusions

We have demonstrated excellent dispersion of MWCNTs in aqueous SDS solutions. The sonication-driven dispersion mechanism and the factors that optimize its efficiency have systematically been studied by using UV–visible spectroscopy and TEM. During sonication, MWCNTs are gradually exfoliated and disentangled from aggregates and bundles and stabilized by SDS. The maximum achievable degree of dispersion of MWCNTs is reached after a certain sonication energy, which is monitored by UV–vis in a simple manner. The SDS molecules are adsorbed on the surface of MWCNTs and prevent re-aggregation, and the colloidal stability of aqueous MWCNT dispersions is maintained for several months. The minimum SDS to MWCNT ratio of 1.5–1 by weight is required to achieve dispersions with maximum achievable dispersion of MWCNTs in the aqueous SDS solution. More energy is necessary to disperse higher concentrations of MWCNTs. The maximum concentration of MWCNTs that can be homogeneously dispersed is about 1.4 wt%; for higher concentrations of MWCNTs, and accordingly of SDS, no valuable dispersion and stabilization of the MWCNT in the aqueous solution is achieved, most probably by the depletion-induced aggregation of the MWCNTs.

The presented results allow control of the dispersion and stabilization process of MWCNTs in aqueous SDS solution. Moreover, to prevent MWCNT rupture and guarantee maximum dispersion, the minimum sonication energy required and the optimum MWCNTs and SDS concentration can be chosen for certain applications, merely based on simple UV–vis measurements. We believe that our results, at least the general trends, are relevant for sonication-driven dispersion of SWCNTs in aqueous SDS solutions as well.

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